

Business Logic Vulnerability – Unauthorized Refund Manipulation in crAPI :

Overview

During security testing of the intentionally vulnerable **crAPI** application, I identified a business logic vulnerability in the product return workflow.

A normal authenticated user was able to manipulate order-related parameters and receive a refund significantly greater than the amount originally paid.

By modifying the product quantity from **1 to 30** and changing the order status from Pending to Returned, the application processed a **\$300 refund** for a product that originally cost **\$10**.

Testing Environment: crAPI deliberately vulnerable API lab

Authorization: Educational and authorized lab testing only

Vulnerability Details

Vulnerability Type: Business Logic Vulnerability

Related Weakness: Improper Server-Side Validation / Unauthorized Modification of Sensitive Properties

Potential OWASP API Category: API3:2023 – Broken Object Property Level Authorization

Impact

An authenticated low-privileged user could manipulate the return workflow to receive an unauthorized refund.

In this test:

- Original account balance: **\$100**
- Product price: **\$10**
- Purchased quantity: **1**
- Balance after purchase: **\$90**
- Manipulated return quantity: **30**
- Calculated refund: **\$300**
- Final account balance: **\$390**

This resulted in an unauthorized financial gain of **\$300** beyond the legitimate \$10 refund.

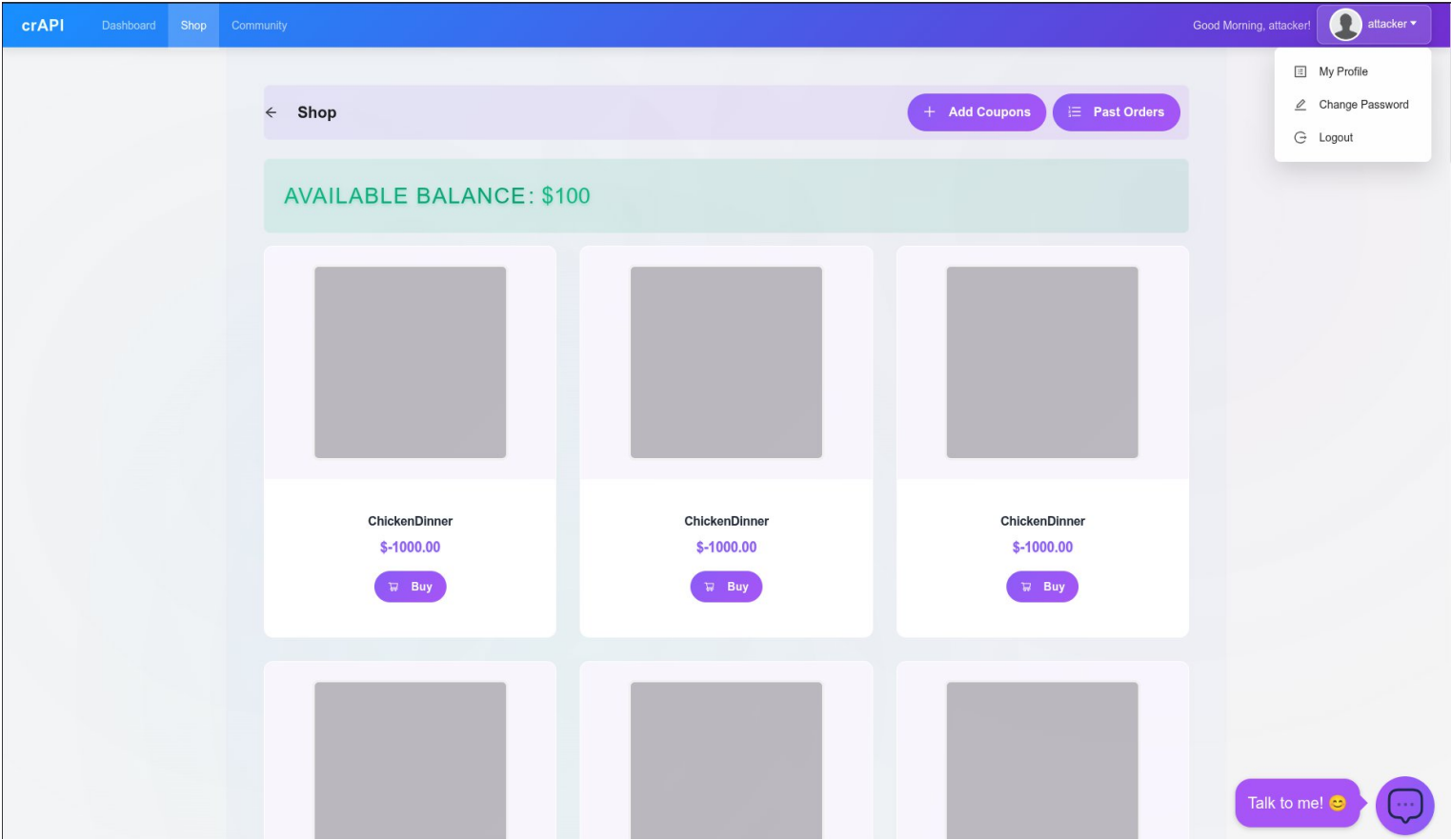
Attack Scnerio :

1) Normal User Account :

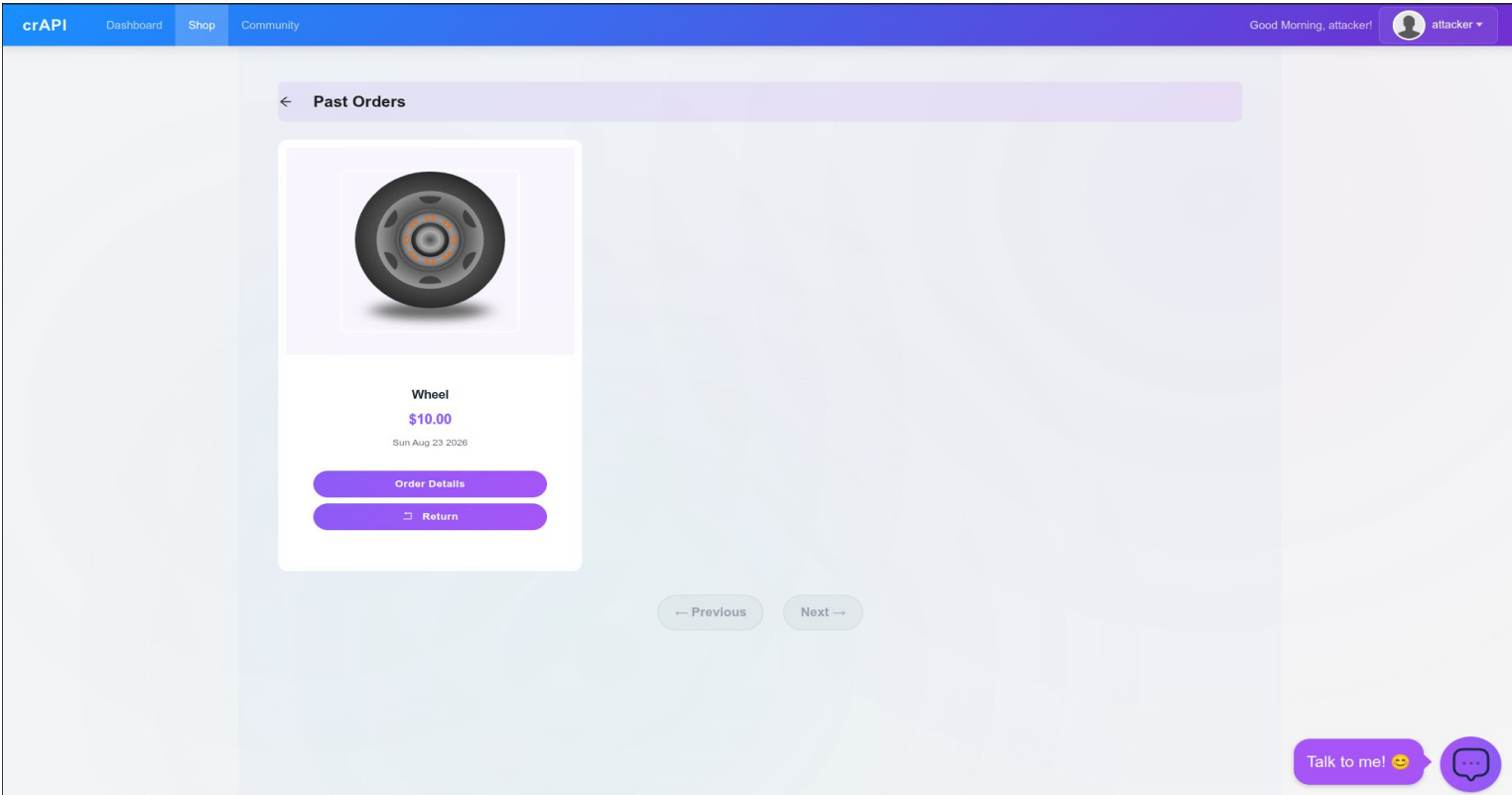
I registered and logged into crAPI as a normal authenticated user.

The account initially contained:

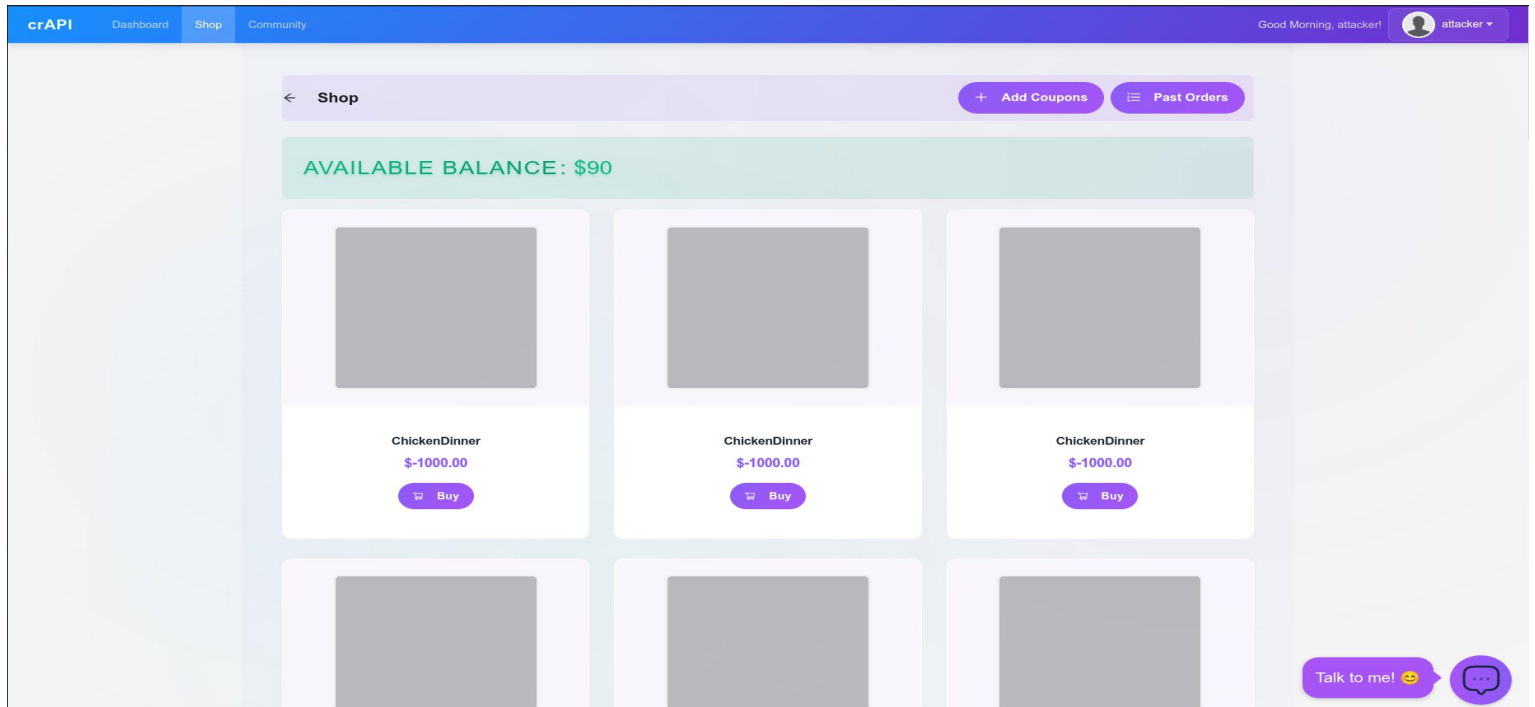
Available Balance: \$100



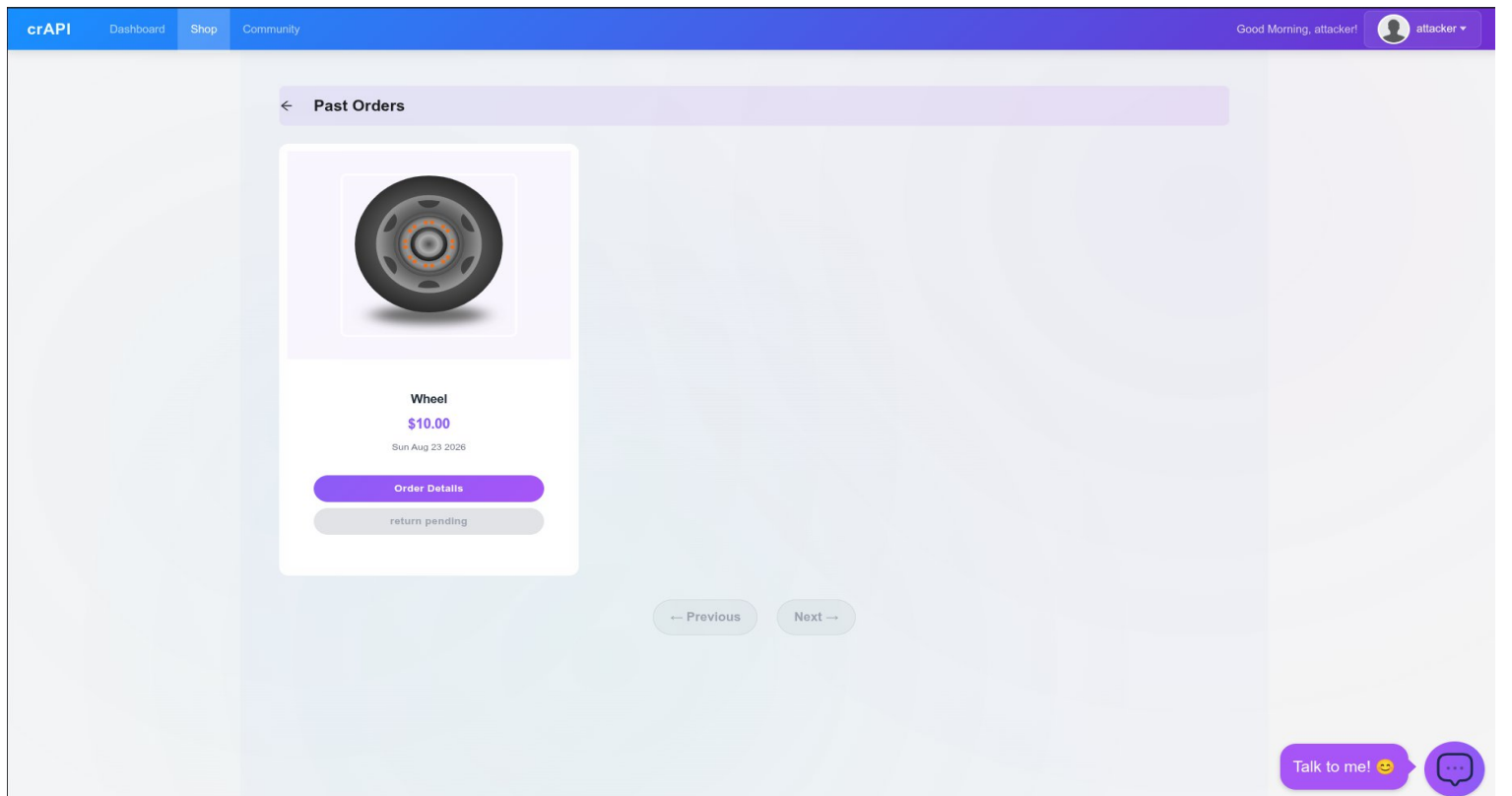
2) Purchase a product with a price \$10



After purchase a product account balance is \$90



3) Initiate the return process



4) Captured Order Details request :

[illegible]

5) Manipulate sensitive properties

[illegible]

Change the Quantity 1 to 30

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Burp

Project

Intruder

Repeater

View

Help

Param Miner

Burp Suite Community Edition v2026.7.2 - Temporary Project

Dashboard

Target

Proxy

Intruder

Repeater

Collaborator

Sequencer

Decoder

Comparer

Logger

Organizer

Extensions

Discover

Request Timer

Attack Surface Detector

Autorize

JWT Editor

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Cancel

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Burp AI

Target: http://crapi.local:8888

HTTP/1

Request

Response

Inspector

Pretty

Raw

Hex

JSON Web Token

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The backend processed the manipulated quantity and calculated the refund based on **30 products** rather than the **1 product actually purchased**.The manipulated return was accepted, and the account received a **\$300 refund**.

Root Cause :

The vulnerability appears to occur because the server trusted client-controlled order properties during the return and refund workflow.

The backend should not rely on values such as:

- Product quantity
- Product price
- Refund amount
- Order status

when those values are supplied or modified by the client.Instead, the server should retrieve the original order information from a trusted server-side data source and validate the return against that information.

Security Impact :

A malicious authenticated user could potentially:

- Manipulate return quantities
- Obtain refunds exceeding the original purchase amount
- Increase their account balance without authorization
- Abuse weaknesses in financial business workflows

In a real-world e-commerce application, this type of flaw could lead to **direct financial loss**.

Remediation :

- I. Validate return quantities server-side
- II. Calculate refunds using trusted server-side data
- III. Protect sensitive order properties
- IV. Enforce valid state transitions
- V. Validate the complete transaction

Key Takeaway :

This finding demonstrates why security testing should not focus only on traditional vulnerabilities such as SQL injection or authentication bypasses.

Even when an API correctly authenticates a user, insecure business rules and insufficient server-side validation can allow an attacker to abuse legitimate functionality for unauthorized financial gain.

Tested only in the intentionally vulnerable crAPI lab environment for educational and authorized security research.